Statistics Cheat Sheet

Data, distributions, and visual interpretation

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This sheet follows the assigned sections of *Introduction to Modern Statistics (2e)* (IMS) (Çetinkaya-Rundel and Hardin 2024). The textbook links below work online and in the PDF.

> The big idea
>
> **Start with the data, not the chart.**
> Before visualizing anything, ask:
> 1. **What does one row represent?**
> 2. **What variables do I have?**
> 3. **What type is each variable?**
> 4. **Am I describing one variable or a relationship?**
> 5. **What population do I ultimately care about?**

1. Data basics

Observations, variables, and data frames

Textbook: IMS § 1.2.1.

Most datasets can be imagined as a table:

artwork	medium	viewing_time	likes
A	print	18.4	12
B	digital	7.9	34
C	print	12.1	19

- **Observation / case / observational unit:** one thing being studied.
  - Usually one **row**.
  - Here: one artwork.
- **Variable:** a characteristic measured for every observation.
  - Usually one **column**.
  - Here: `medium`, `viewing_time`, `likes`.
- **Value:** the particular measurement in one cell.
  - e.g. `18.4` seconds.

A useful question is:

> **What does one row mean?**

If you cannot answer that clearly, you probably do not yet understand the dataset.

Tidy data

A common structure is **tidy data**:

- one row = one observation
- one column = one variable
- one cell = one value

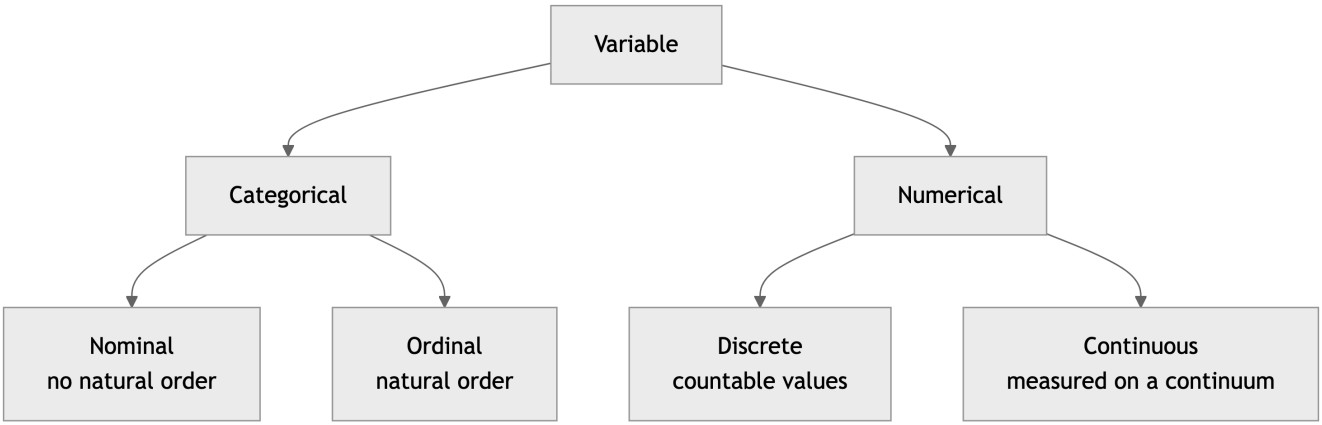
This structure makes data easier to summarize, visualize, and analyze.

> ⚠ Design-data trap
>
> A spreadsheet that *looks* neat to a human is not necessarily tidy data.
> For example, using separate columns for `Monday`, `Tuesday`, `Wednesday`, etc. may encode values in column names instead of storing **day** as a variable.

Types of variables

Textbook: IMS § 1.2.2.

The most important split is:



Numerical variables

Values represent quantities. Differences between them, and summaries such as averages, can be meaningful.

Examples:

- viewing time
- font size
- age
- screen width
- number of visitors
- number of clicks

Discrete numerical

Values come in distinct steps, often because they are **counts**.

Examples:

- 0, 1, 2, 3 clicks
- number of exhibition visitors
- number of colors used

You cannot normally observe 3.72 visitors.

Continuous numerical

Can theoretically take any value within some interval.

Examples:

- reading time: 4.81 s
- height: 172.3 cm
- temperature
- distance

Measurements may be rounded, but the underlying quantity is continuous.

Categorical variables
Values represent groups or labels.
Examples:

- medium: `print`, `digital`, `installation`
- operating system
- country
- typeface family
- yes/no

> ! Numbers are not automatically numerical data
>
> A student ID, postcode, telephone number, or room number may contain digits, but averaging them makes no sense.
> Ask:
>
> **Does arithmetic on these values mean anything?**
>
> If not, treat the variable as categorical.

Nominal vs. ordinal
Nominal
Categories have no inherent ranking.
`medium = print / web / installation`
Ordinal
Categories have a meaningful order.
`satisfaction = poor / fair / good / excellent`
But the *distance* between ordinal categories is not necessarily equal.
`poor → fair`
does not necessarily represent the same change as
`good → excellent`.

Relationships between variables

Textbook: IMS § 1.2.3.

Often the goal is not to describe one variable but to ask whether **two variables are related**.

Design examples:

- Does larger text correspond to shorter reading time?
- Do visitors spend more time with larger artworks?
- Is device type related to task completion?
- Does color contrast correspond to higher click-through rates?

Two variables are **associated** if values of one variable tend to correspond systematically with values of another.

Positive association

As one variable increases, the other tends to increase.

`screen size ↑ → time watching video ↑`

Negative association

As one increases, the other tends to decrease.

`font size ↑ → reading errors ↓`

No evident association

Knowing one variable gives little information about the other.

> ⚠ Association ≠ causation
>
> A visible relationship does **not** automatically mean that one variable caused the other.
>
> There might be:
>
> - other variables involved;
> - selection effects;
> - common causes;
> - coincidence.
>
> A plot can show an **association**. Establishing causality requires an appropriate study design, typically an experiment.

Explanatory and response variables

Textbook: IMS § 1.2.4.

When the research question suggests that one variable may explain or predict another:

Explanatory variable → Response variable

Example:

> Does increasing line spacing improve reading speed?

- **Explanatory:** line spacing
- **Response:** reading speed

In a scatterplot, the convention is usually:

- explanatory variable → **x-axis**
- response variable → **y-axis**

But assigning these labels does **not** prove causality.

2. Population and sample

Textbook: IMS § 2.1.1.

Population

The **population** is the complete group you ultimately want to understand.
Example research question:

> What is the average time visitors spend in this museum exhibition?

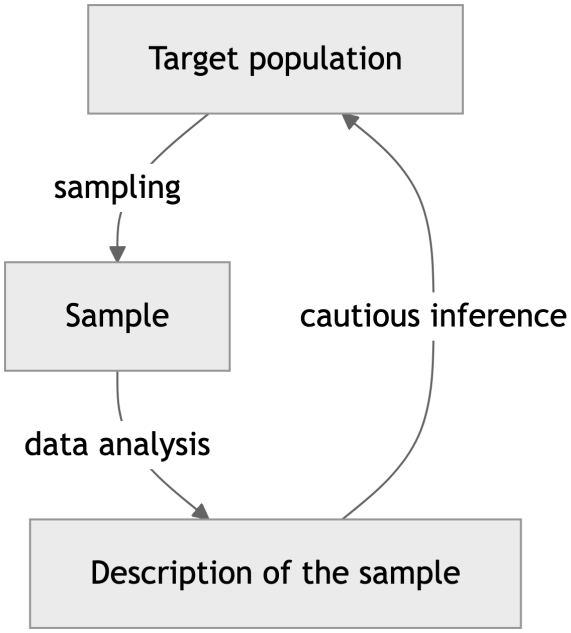
Possible population:

> All visitors to the exhibition during its entire run.

Sample

The **sample** is the set of observations you actually collect.
Example:

> 180 visitors observed during four selected days.



Census

A **census** measures every case in the population.
This is often:

- expensive;
- slow;
- impractical;
- impossible.

So we usually study a sample instead.

> ! The sample is not automatically the population
>
> Observing 100 art students does not automatically tell you about **all adults**.
> When reading a statistic, ask:
>
> **Who was actually measured, and who is the conclusion supposed to describe?**

---

3. Categorical data: contingency tables & bar plots

Textbook: IMS § 4.1.

Contingency table

Use a **contingency table** to summarize **two categorical variables**.
Example: device type and task completion.

	Completed	Not completed	Total
Mobile	42	18	60
Desktop	32	8	40
Total	**74**	**26**	**100**

Each inner cell describes one **combination** of categories.
For example:

> 18 mobile users did not complete the task.

The final row and column are **marginal totals**.

What to look for
Ask whether the distribution of one variable changes depending on the other.
For example:

> Is task completion distributed similarly for mobile and desktop users?

The two groups hold different numbers of users, so raw counts can mislead. Convert each row to a proportion of its own total:

```
Mobile completion:   42 / 60 = 70%
Desktop completion:  32 / 40 = 80%
```

> ! Compare proportions, not only counts
>
> When the group sizes differ, compare proportions within each group. Mobile shows more completions than desktop, 42 against 32, but a smaller share of its own users, 70% against 80%.

A contingency table gives the raw material for studying relationships between categorical variables.

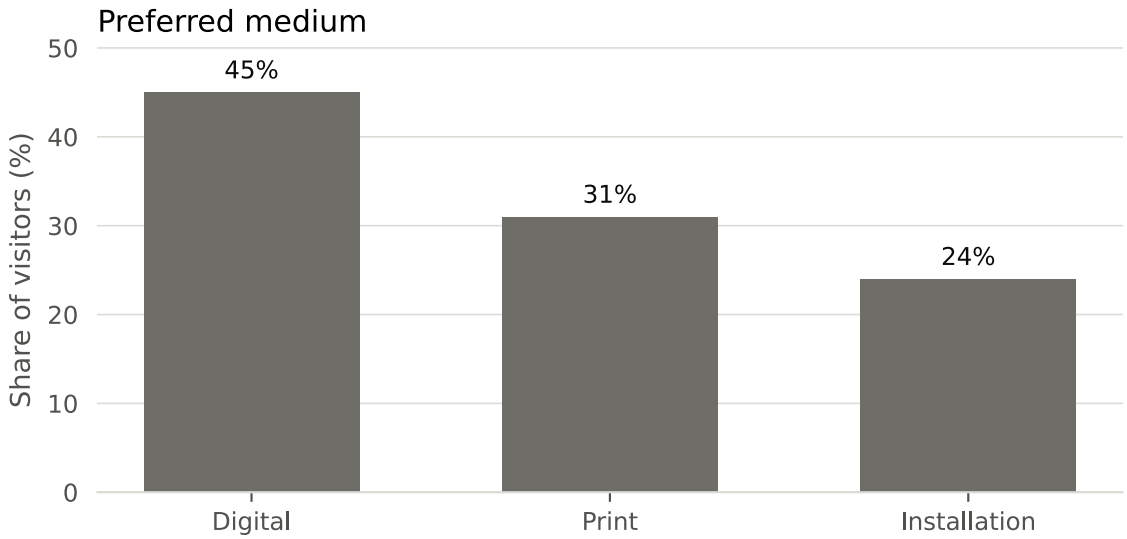
Bar plot

Use a bar plot for a **categorical variable**.

Bar height can represent:

- **count**, or
- **proportion / percentage**.

Example:



Good bar-chart practice

- Start the quantitative axis at zero when bar **length** represents magnitude.
- Label categories clearly.
- Sort intentionally:
  - natural order;
  - ascending/descending value;
  - meaningful conceptual order.
- Avoid decoration that competes with comparison.
- Use counts when the absolute number matters.
- Use proportions when the share of observations matters.

4. Scatterplots for paired data

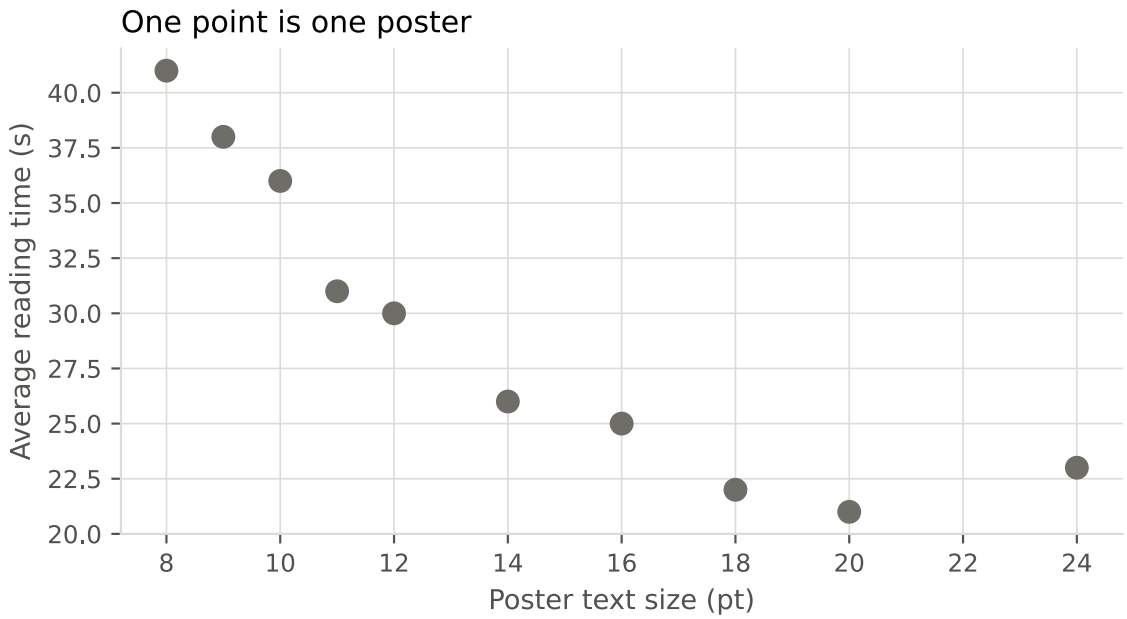
Textbook: IMS § 5.1.

Use a **scatterplot** when you have **two numerical variables measured on the same observations**.

Each point = one case.

Example:

- x = poster text size
- y = average reading time
- one point = one poster



Read a scatterplot using four questions

1. Direction

Does y generally:

- increase as x increases? → **positive**
- decrease as x increases? → **negative**
- show no clear direction?

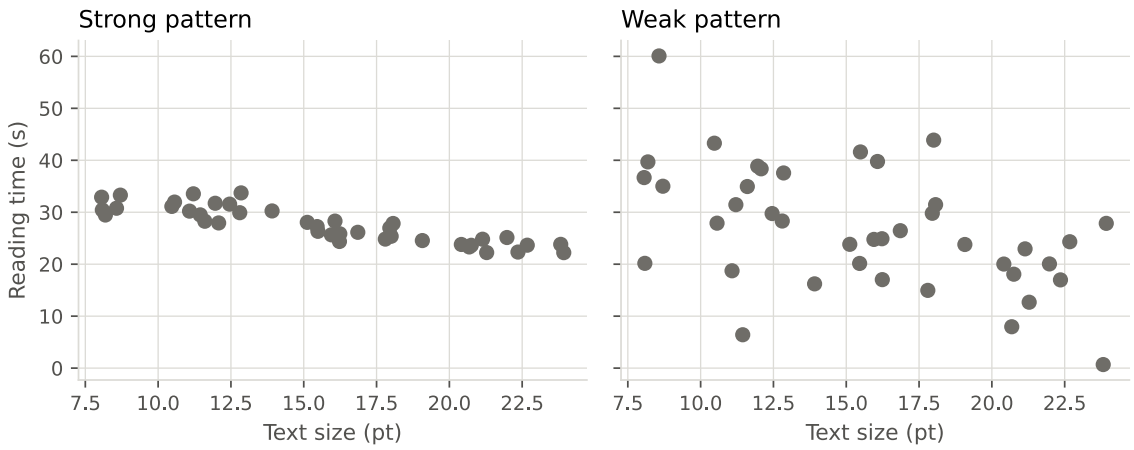
2. Form

Is the relationship:

- approximately linear?
- curved / nonlinear?
- clustered?
- something more complicated?

Do not assume every relationship should be a straight line.

3. Strength
How closely do the points follow the overall pattern?
Compare. Both panels share the same underlying relationship. Only the scatter around it differs.



4. Unusual observations
Look for points far from the main pattern.
They may represent:
- a genuine exceptional case;
- measurement error;
- data-entry error;
- a subgroup worth investigating.

Do not silently delete them.

> ⚠ Scatterplot ≠ proof
>
> Even a beautiful, strong relationship only establishes an **association** from the plot itself.
> It does not establish that changing x will cause y to change.

Sentence template

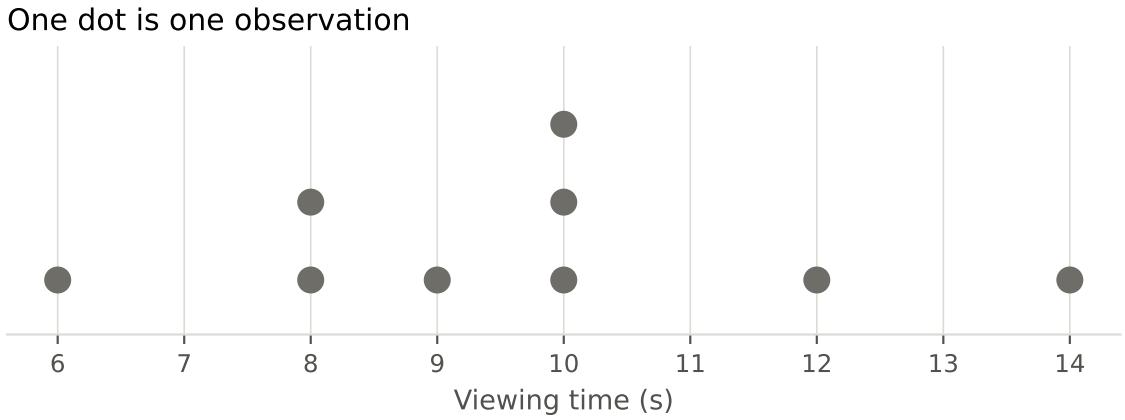
> As **[x]** increases, **[y]** tends to **[increase/decrease]**. The relationship appears **[roughly linear/nonlinear]**, with **[little/substantial]** scatter and **[no/some]** unusual observations.

5. Dot plots and the mean

Textbook: IMS § 5.2.

Dot plot

A dot plot displays the distribution of **one numerical variable**.
Each dot represents an observation.
It works especially well for relatively small datasets because you can still see individual values.



For large datasets, histograms are usually more readable.

Mean

The **mean** is the arithmetic average and a measure of the center of a distribution.
For observations

$$x_1, x_2, ..., x_n$$

the **sample mean** is

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Where:

- x_i = one observed value
- n = number of observations
- $\bar{x}$ = sample mean

Example

Five reading times:

$$8,9,10,13,15$$

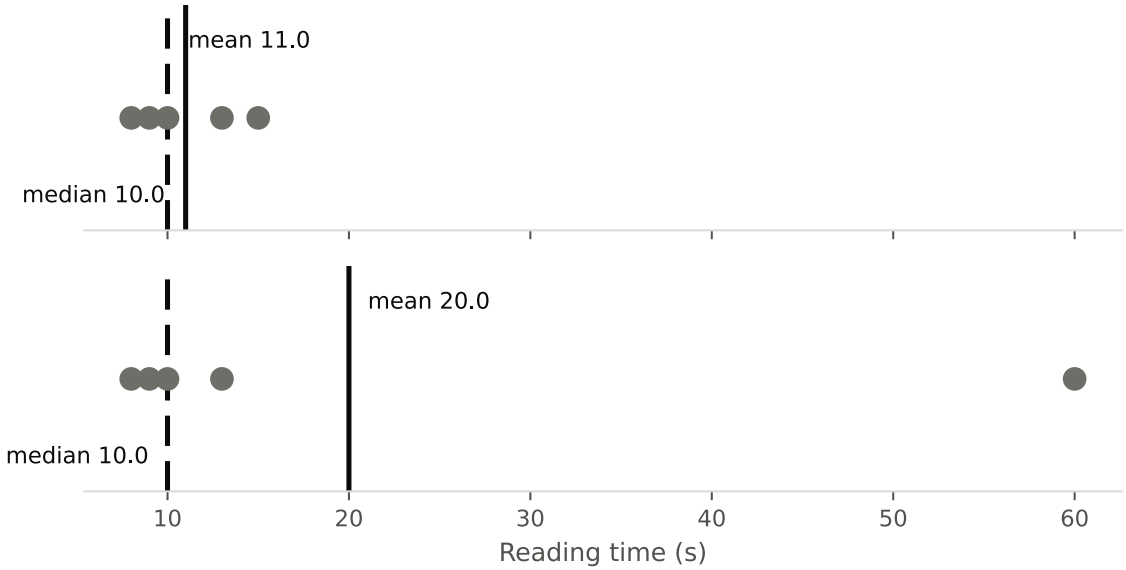
Then

$$\bar{x} = \frac{8 + 9 + 10 + 13 + 15}{5} = 11.$$

Average reading time = **11 seconds**.

Think of the mean as a balancing point

The mean is not simply “the most common” or “the middle” observation.
It can be imagined as the **balance point** of the distribution.
The panels below hold the same four small values. Only the last value changes. The mean follows it. The median does not.



Values far away from the center contribute strongly to where this balancing point lies.

Sample mean vs. population mean

Sample mean:

$$\bar{x}$$

Population mean:

$$\mu$$

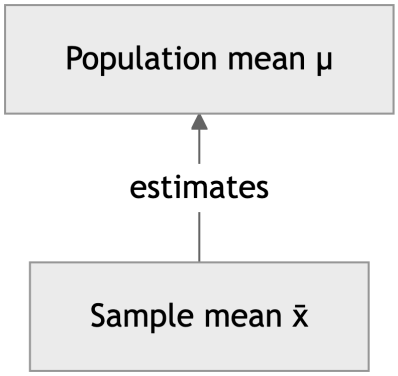
Usually we cannot measure every case in the population, so we use

$$\bar{x}$$

as an estimate of

$$\mu.$$

This step needs an appropriate sampling design. A sample that is not representative gives a sample mean that can sit far from the population mean, however large the sample is.



Why means are useful

Means put totals into a common unit.
Instead of:

> 220 minutes of viewing across 20 visitors

we can say:

$$\frac{220 \ \text{min}}{20 \ \text{visitors}} = 11 \ \ \text{min/visitor} \ .$$

This makes comparisons easier.

Weighted means

Be careful when averaging averages from groups of different sizes.
Suppose:

Test group	Users	Mean completion time
A	10	20 s
B	90	30 s

The overall mean is **not**

$$\frac{20 + 30}{2} = 25.$$

Group B represents nine times as many users.
Instead:

$$\bar{x} = \frac{10(20) + 90(30)}{100} = 29 \ \ \text{s} \ .$$

> ⚠ Common mean mistake
>
> Do not give groups equal weight when the groups contain very different numbers of observations.
> Ask what each number being averaged actually represents.

6. Histograms and distribution shape

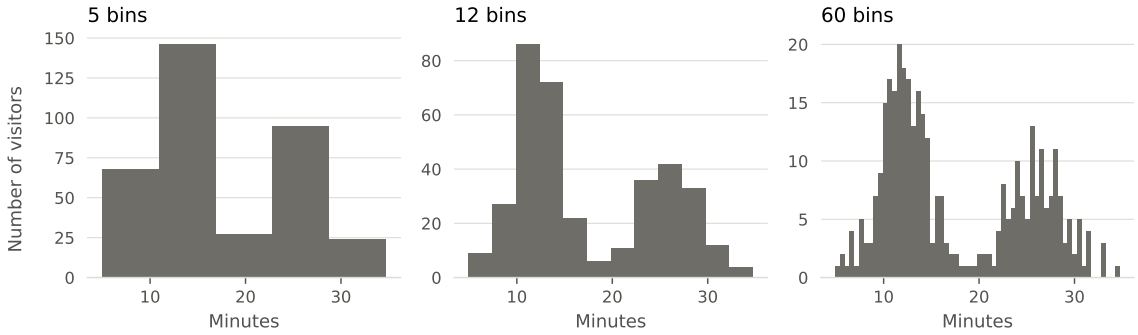
Textbook: IMS § 5.3.

Use a **histogram** to display the distribution of **one numerical variable**, especially when there are too many observations for a dot plot.

Instead of drawing each observation separately, values are grouped into intervals called **bins**.

For equal-width bins, higher bars mean more observations occur in that numerical region.

The three panels below show **the same 360 visitors**. Only the number of bins changes. Five bins flatten the detail. Sixty bins show noise between the bins as if it were structure. Twelve bins show the two groups of visitors most clearly.



> ⚠ The bin width is a design decision
>
> Nobody measured the bins. Someone chose them. A different choice tells a different story about the same observations.

What a histogram preserves

It helps reveal:

- where observations concentrate;
- overall shape;
- skew;
- peaks / modes;
- approximate center;
- approximate spread;
- unusual regions or gaps.

What it hides

Because observations are grouped into bins, you cannot usually recover the exact individual values.

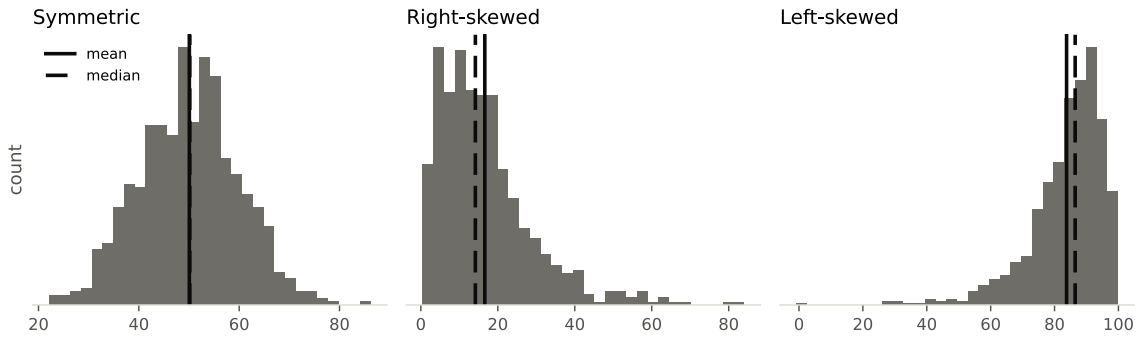
A histogram is therefore a **summary of the distribution**, not a case-by-case display.

Shape: symmetric vs. skewed

Symmetric
Both sides trail off in roughly similar ways.

Right-skewed
Long tail points toward **higher values**.
Often:
- many observations are relatively low;
- a few observations are very high.

Left-skewed
Long tail points toward **lower values**.
Compare the three shapes. The solid line marks the mean. The dashed line marks the median. Skew pulls the mean into the tail and leaves the median behind.



! Name the skew from the tail

Right-skewed does not mean “most observations are on the right.”
It means the **long tail points right**.
Likewise, left-skewed = long tail to the left.

Modes

A **mode** is a prominent peak in a distribution.
The useful question is not necessarily:

> Which exact value occurs most often?

For numerical data, instead ask:

> How many **prominent peaks** are visible?

- **unimodal** → one prominent peak
- **bimodal** → two
- **multimodal** → more than two

A bimodal distribution may suggest that two different groups have been combined.
Example:

> Heights measured from children and adults might produce two peaks.

Description template

> The distribution of **[variable]** is **[unimodal/bimodal/multimodal]** and **[symmetric/right-skewed/left-skewed]**, with most observations around **[region]** and **[any unusual features]**.

7. Box plots, quartiles, and the median

Textbook: IMS § 5.5.

Median

First order the observations from smallest to largest.
The **median** is the value in the middle.

Odd number of observations

```
2  4  7  8  19
      ↑
   median
```

Median = **7**

Even number of observations

```
2  4  7  8  19  23
      ↑  ↑
```

Average the two middle values:

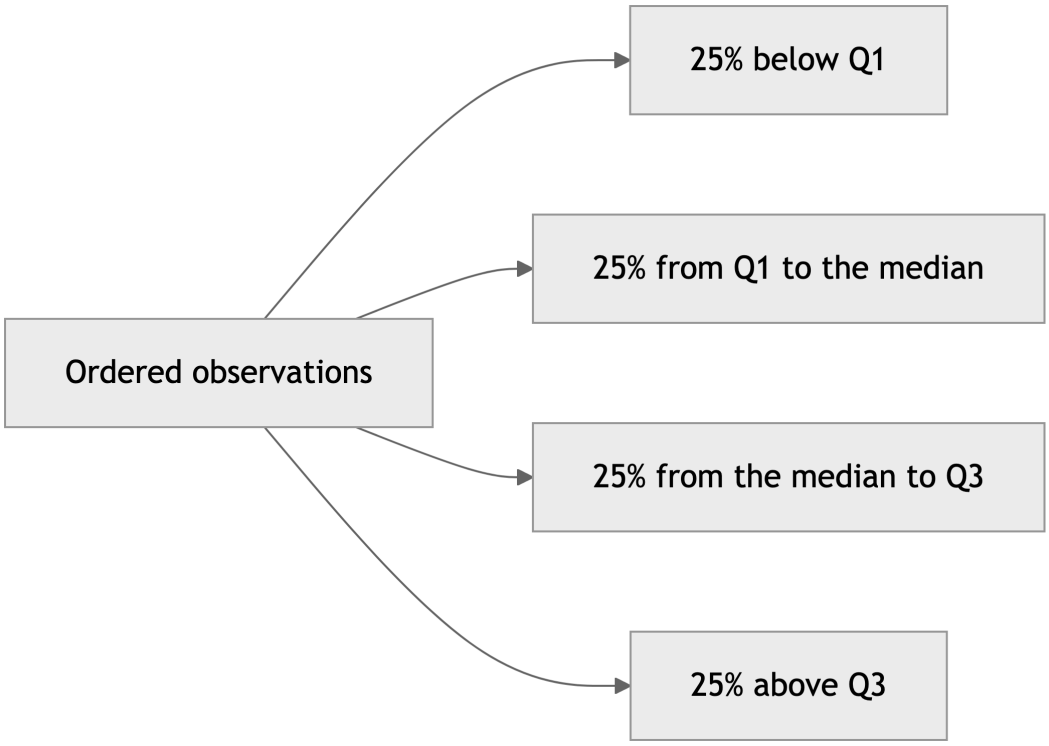
$$\text{median} = \frac{7+8}{2} = 7.5$$

The median is also the **50th percentile**. It splits the ordered observations into a lower half and an upper half.

Quartiles

Quartiles split an ordered distribution into four parts.

- Q_1 = 25th percentile
- median = 50th percentile
- Q_3 = 75th percentile



Approximately:

- 25% below Q_1
- 25% between Q_1 and median
- 25% between median and Q_3
- 25% above Q_3

Interquartile range

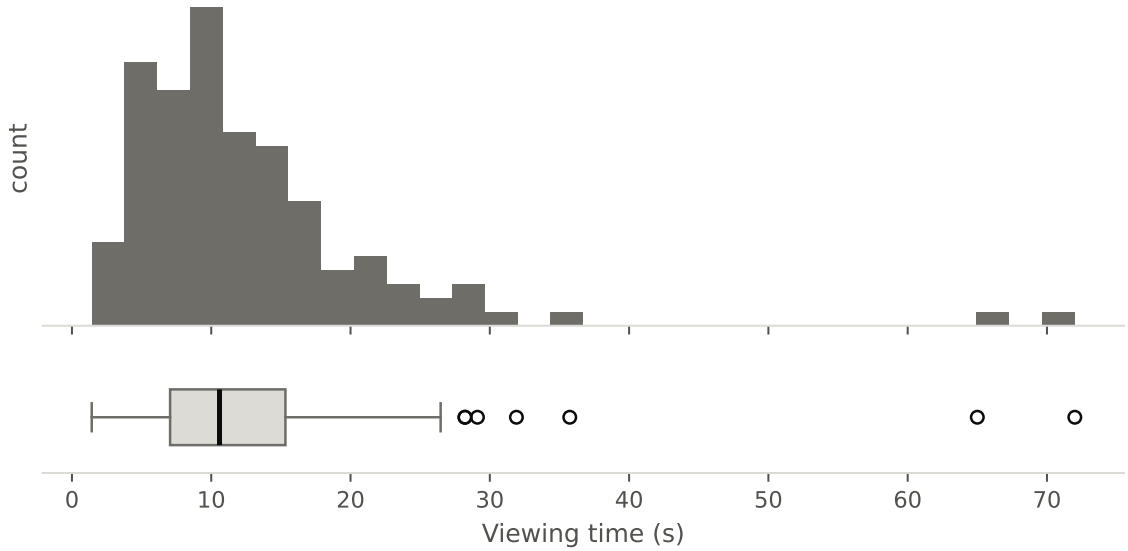
The **interquartile range (IQR)** measures the width of the middle 50%:

$$IQR = Q_3 - Q_1.$$

Large IQR → middle half of observations are spread widely.
Small IQR → middle half are relatively concentrated.

Reading a box plot

The two panels below show the same observations. The histogram shows the shape. The box plot compresses the distribution into quartiles, a median, whiskers, and potential outliers.



The black line is the median. The open circles beyond the whisker are potential outliers.
The box runs from:

$$Q_1 \to Q_3$$

and therefore contains the **middle 50%** of observations.
The line inside the box is the **median**.
Each whisker reaches the most extreme observation that still lies within 1.5 × IQR of the box. Observations beyond those fences are drawn individually as potential outliers.

Potential outliers: the 1.5 × IQR rule

A commonly used rule marks an observation as a potential outlier when it lies below

$$Q_1 - 1.5(IQR)$$

or above

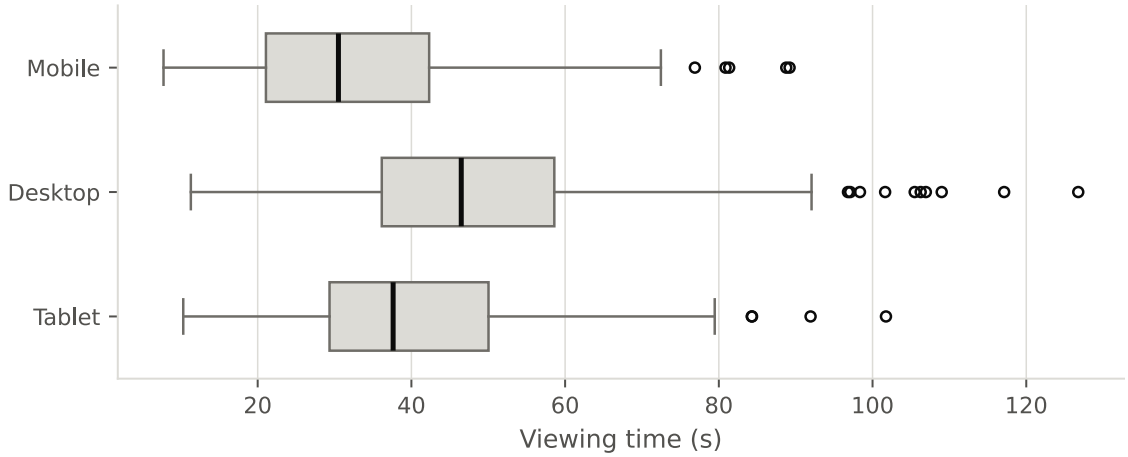
$$Q_3 + 1.5(IQR).$$

These boundaries are sometimes called **fences**.

> ⚠ Outlier does not mean error
>
> A point beyond an outlier fence may be:
> - an error;
> - an unusual but genuine case;
> - evidence of skew;
> - the most interesting observation in the dataset.
>
> Investigate it before deciding what to do.

Compare groups with side-by-side box plots

Draw one box plot for each category, on a shared numerical axis. This is the standard display for **one categorical and one numerical variable**.



Read it by comparing the medians, then the boxes, then the whiskers and the outliers.

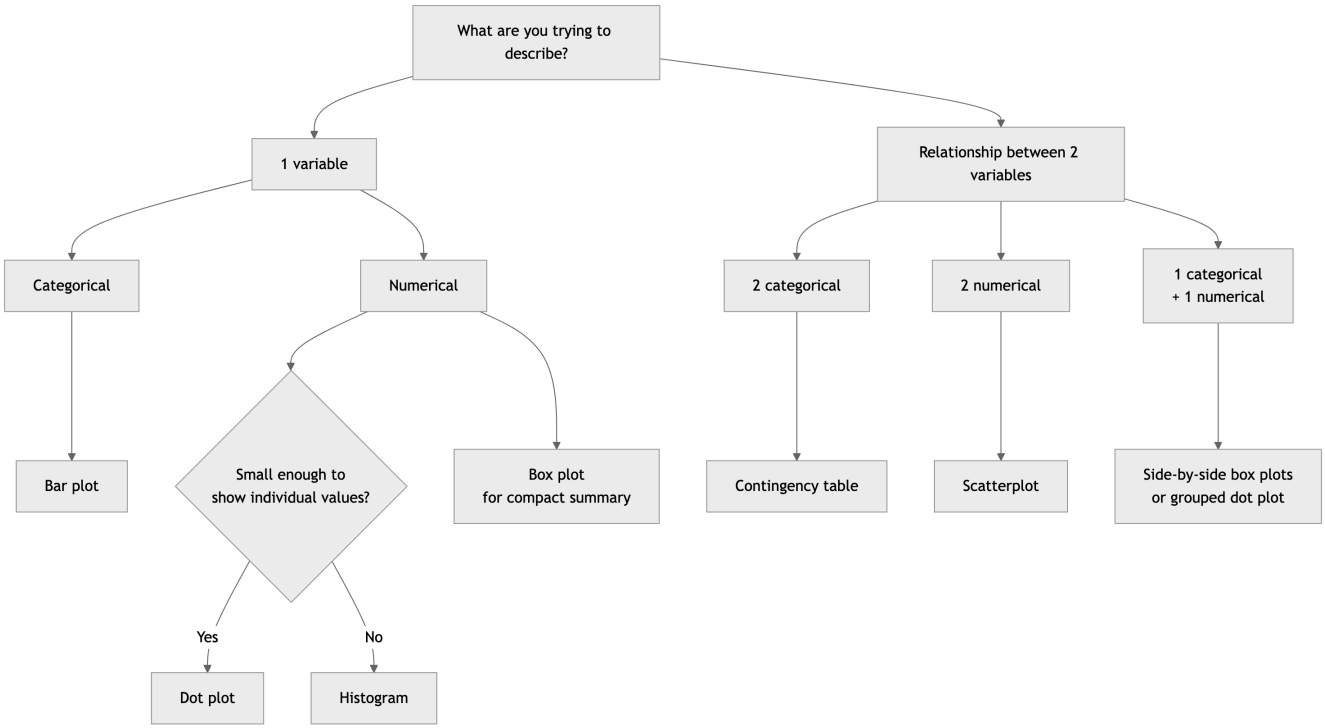
A grouped dot plot works as well when each group holds few observations, because it still shows the individual values.

Bar plot vs. histogram

These are easy to confuse.

	Bar plot	Histogram
Variable	Categorical	Numerical
Each bar represents	A category	A numerical interval/bin
x-axis	Labels	Numerical scale
Order	Can often be rearranged	Must follow numerical order
Gaps	Usually separated	Usually adjacent
Main question	How common is each category?	What is the shape of the distribution?

Which display should I use?



For one categorical and one numerical variable, ask:

> **How does the numerical distribution differ across the categories?**

For example:

- Does viewing time differ by device type?
- Does completion time differ across task categories?

Fast interpretation checklist

When you see a visualization, ask:

1. **What is one observation?**
2. **Which variables are shown?**
3. **Are they categorical or numerical?**
4. **What is the population, and what is the sample?**
5. **Is the visualization appropriate for the variable type?**
6. **Where is the center?**
7. **How spread out are the observations?**
8. **What is the shape?**
9. **Are there clusters, gaps, or unusual observations?**
10. **Is the chart showing association, or is someone claiming causation?**

Five formulas, symbols, and definitions worth remembering

Sample mean

$$\boxed{\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i}$$

Population mean

$$\boxed{\mu}$$

Median

$$\boxed{50\text{th percentile}}$$

Interquartile range

$$\boxed{IQR = Q_3 - Q_1}$$

Common potential-outlier fences

$$\boxed{Q_1 - 1.5IQR} \qquad \boxed{Q_3 + 1.5IQR}$$

One-minute summary

Situation	Main idea
Dataset	rows = observations; columns = variables
Numerical variable	values are quantities; differences and averages can be meaningful
Categorical variable	values represent groups
Discrete	countable numerical values
Continuous	measured along a continuum
Population	full group you want to understand
Sample	observations you actually have
Two categorical variables	contingency table
One categorical + one numerical variable	side-by-side box plots
One categorical variable	bar plot
Two numerical variables	scatterplot
Small numerical dataset	dot plot
Numerical distribution	histogram
Arithmetic center	mean $\bar{x}$
Middle observation	median
Middle 50%	Q_1 to Q_3
Middle-50% spread	IQR
Compact distribution summary	box plot
Long tail right	right-skewed
Long tail left	left-skewed
Visible relationship	association
Association	**not automatically causation**

For information designers

A visualization is not merely decoration applied after analysis.
Choosing between a bar plot, histogram, scatterplot, or box plot is part of the **reasoning**: each graphic answers a different kind of question and preserves or hides different features of the data.